CASE 34
SYSTEMIC JUVENILE IDIOPATHIC ARTHRITIS

Cytokine dysregulation leading to inflammatory disease.

Cytokines are small proteins that act over short distances to affect adjacent cells, or they can be carried in the bloodstream over large distances, affecting the entire body. The pro-inflammatory cytokines, such as interleukin-1β (IL-1β), tumor necrosis factor-α (TNF-α), and IL-6, elicit powerful inflammatory effects that lead to aberrant thermoregulation, metabolism, hematopoiesis, and tissue inflammation (Fig. 34.1).

IL-1β is a prototypic pro-inflammatory cytokine. Like many other cytokines, it is synthesized as an inactive precursor molecule, and cleavage of pro-IL-1β by caspase 1 is required for its activation and secretion (see Fig. 33.4). Its production and release are regulated at multiple levels, including this cleavage step. The activation of caspase 1 is dependent on the inflammasome, a multiprotein complex activated by diverse substances such as microbial constituents, vaccine adjuvants, pollutants, amyloid, and uric acid crystals. Several monogenic autoinflammatory disorders have been identified in which genes encoding inflammasome proteins are mutated. These disorders are characterized by recurrent fevers, elevation of inflammatory markers, and skin, joint, and organ inflammation (see, for example, Case 33).

Another potent inflammatory cytokine is IL-6. IL-6 is produced primarily by T cells and is important for T-cell differentiation and proliferation as well as the differentiation of B cells into plasma cells. A subset of helper T cells, T_H17 cells, produces IL-17 and TNF-α, as well as IL-6. T_H17 cells have been shown to have an important role in inflammation and in the induction of autoimmune diseases. Along with lymphocyte-specific effects, excess IL-6 induces symptoms of fever, anorexia, and fatigue in patients, as well as elevation of acute-phase reactants (e.g., C-reactive protein (CRP), serum amyloid A, and fibrinogen).

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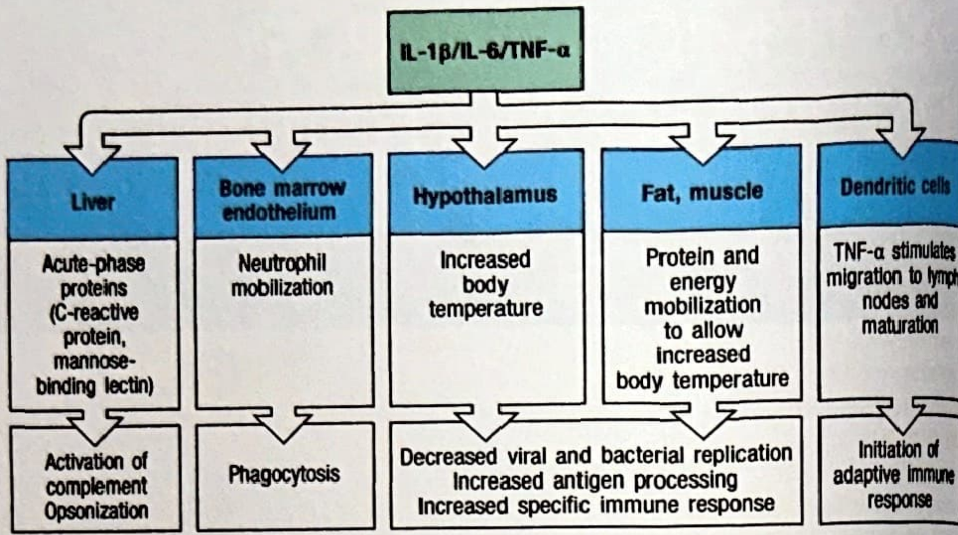
TOPICS BEARING ON THIS CASE:

- Cytokines and the inflammatory response
- Cytokine processing
- Acute-phase responses
- T_H17 cells
- Anti-cytokine therapeutic agents
- The autoinflammatory syndromes

Fig. 34.1 Pro-inflammatory cytokines. IL-1β, IL-6, and TNF-α are powerful pro-inflammatory cytokines. They elicit responses not only from immune cells but also from various target organ systems. These cytokines promote a catabolic state with breakdown of muscle and fat, along with fever and an increased production of acute-phase proteins.



The case of Catherine Earnshaw: the fever that wouldn't stop.

8-year-old girl with at least 2 weeks of fever, rash, joint swelling, and pain.

Catherine had enjoyed normal good health until she was 8 years old, when she developed recurrent high fevers and fatigue. She also found it difficult to get out of bed in the morning—she ached all over. Her parents noticed that a pink rash on her shoulders and arms appeared during the fevers. After more than a week of these symptoms, Catherine's parents were concerned that this was more than a typical childhood illness and took her to the pediatrician. Although Catherine had no fever or rash at the time, Dr Brontë noticed subtle swelling of several joints, along with pain on movement; she initially suspected a parvovirus infection. Parvoviral serologies were sent, and ibuprofen and acetaminophen were recommended for symptomatic relief. A few days later, Catherine's parvoviral serologies returned negative.

Parvovirus serologies negative.

Catherine went back to the pediatrician after another week of twice-daily fevers, joint pain, and rash. Laboratory studies were sent at that visit. Catherine's white blood cell count was 12,500 μl^{-1} (high), and her hemoglobin was 10 g dl^{-1} (low). Erythrocyte sedimentation rate (ESR) was also elevated, at 82 mm h^{-1}. She was sent to the local children's hospital for further evaluation.

Elevated white blood cell count and inflammatory markers (ESR, CRP, and ferritin).

In the hospital, additional laboratory studies were carried out to look for infection, all of which were negative. The hematologists took a peripheral blood smear to exclude hematologic malignancy, which appeared normal apart from a mild anemia and increased neutrophil count. Catherine continued to have intermittent high fevers (Fig. 34.2), and the hospital staff noticed a light pink rash during these times

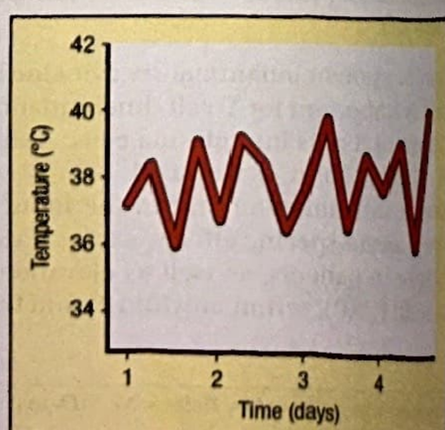


Fig. 34.2 A typical fever curve from a patient with systemic juvenile idiopathic arthritis (sJIA). Patients with sJIA usually have spiking fevers once or twice each day. In between fevers, their body temperature is normal or even low. Patients often seem and feel quite well when they are not feverish. By contrast, joint swelling persists and can cause pain throughout the day.

(Fig. 34.3). **Catherine continued to have difficulty getting out of bed in the morning, but was better able to move later in the day. The rheumatology service saw Catherine and noticed swelling and a decreased range of motion in her ankles, knees, and wrists. Her white blood cell count and ESR continued to be high. Levels of CRP were also very high, at 28 mg dl^{-1}, and ferritin was 1949 ng ml^{-1} (high). Transaminases, electrolytes, creatinine, and immunoglobulins were normal. Tests for anti-nuclear antibodies (ANA) and rheumatoid factor were negative.**

Catherine's symptoms, physical examination, and laboratory results were consistent with systemic juvenile idiopathic arthritis (sJIA). To help control her joint pain and fevers, Catherine was started on the Il-1receptor antagonist, anakinra. Catherine improved significantly and was discharged. She was followed in the rheumatology clinic over the next few months. Catherine did well, but 6 months later, the fevers, rash, and joint pain returned, and she required treatment with a course of prednisone to control her symptoms. She has been doing well since.

Fig. 34.3 The sJIA rash. The classic rash associated with sJIA is a flat, salmon-pink evanescent rash, which tends to appear at the same time as the fever. Photograph courtesy of E. Janssen.

Systemic juvenile idiopathic arthritis.

Frank arthritis on exam significant for sJIA.

Systemic juvenile idiopathic arthritis (sJIA) is a highly inflammatory disease that manifests as arthritis along with prominent systemic features such as high spiking fevers, fatigue, rash, and enlargement of lymph nodes and spleen. It was first described in the 1890s by George Frederick Still, a British pediatrician, and is often referred to as Still's disease. The adult counterpart is termed adult-onset Still's disease (AOSD). The annual incidence of chronic childhood arthritis (JIA) is approximately 1 per 1,000, of whom around 10% develop the sJIA form.

sJIA is often difficult to diagnose. Laboratory tests give nonspecific results, and many of the manifestations, especially rash and fever, are commonly seen with childhood infections. The arthritis sometimes lags behind the onset of fevers by weeks or even months, making correct diagnosis even more difficult. In addition to infection, malignancies such as leukemia and lymphoma enter into the differential diagnosis, because children with these cancers often develop fever, leukocytosis, and bony pain from marrow infiltration. Criteria have been established to classify sJIA (Fig. 34.4). However, these criteria are not a reliable guide to the diagnosis of the patient with new-onset sJIA (~30% sensitive), and many of the exclusion criteria are ignored in clinical practice.

Diagnostic criteria for systemic juvenile idiopathic arthritis
Arthritis + daily fever for 2 weeks or more plus at least one of the following:
Evanescent red rash Generalized lymph-node enlargement Enlargement of the liver or spleen Serositis
Exclusions
Psoriasis in the patient or a first-degree relative Arthritis in an HLA B27-positive male older than 6 years A spondyloarthropathy or acute anterior uveitis in a first-degree relative The presence of IgM rheumatoid factor on two or more occasions at least 3 months apart

Fig. 34.4 Diagnostic criteria for sJIA from the International League of Associations for Rheumatology. sJIA is a clinical diagnosis and a diagnosis of exclusion. These criteria were published in 2001 to aid in the classification of sJIA, and to help distinguish this condition from other childhood febrile diseases for the purposes of research. Pediatric rheumatologists commonly diagnose sJIA in patients who do not meet these criteria, especially at disease onset.

In contrast to rheumatoid arthritis (see Case 35) and other forms of JIA, there is no association with autoantibodies in sJIA. Further, whereas most autoimmune diseases are most prevalent in females, the gender balance in sJIA is nearly even (although slight female preponderance is still probable). In addition, MHC class II associations and autoreactive T cells seen in other forms of arthritis are less striking in sJIA than in other autoimmune arthritides. Overall, sJIA seems to have more in common with the inherited autoinflammatory conditions (see Case 33) than with other types of arthritis. In the active state of the disease, persistent activation of phagocytes and upregulation of inflammatory cytokines are seen. Indeed, polymorphisms in the promoters and coding sequences for the genes encoding TNF-α, IL-6, and macrophage migration inhibitory factor (MIF) have been associated with sJIA. Approximately 10–20% of patients with sJIA develop a "cytokine storm" syndrome termed macrophage activation syndrome, sometimes with a fatal outcome. Many of these patients harbor mutations in genes implicated in perforin-mediated cell–cell killing, implicating aberrant immunological self-control in sJIA pathogenesis. The inciting cause of sJIA remains unknown.

The long-term outcome of sJIA is highly variable. Approximately a third of patients exhibit a monophasic course, ultimately entering long-term drug-free remission, although often after many months of therapy. Almost half will develop a chronic course of, often, intractable chronic arthritis, while the remainder have intermittent recurrences. There is at present no way to predict the course a given patient will take.

Traditionally, treatment of sJIA has begun with nonsteroidal anti-inflammatory medications along with corticosteroids. While some patients do well with this treatment, more than 60% require steroids for more than 6 months, and the likelihood of developing chronic arthritis may not be diminished. Inhibitors of TNF-α (see Case 35) have been used therapeutically in sJIA since the 1990s. Unfortunately, the response rate to these medications is significantly lower than in other types of JIA, and they are largely ineffective for the systemic (fever, rash, organomegaly) manifestations of this disease.

More recently, therapies relying on the inhibition of IL-1 and IL-6 have undergone clinical trials. Increased levels of IL-6 have been observed in the serum and joint fluid of patients with sJIA, and these levels correlate with disease activity. Spikes in IL-6 levels also parallel the fever curve seen in sJIA. Tocilizumab is a humanized monoclonal antibody directed against the IL-6 receptor that has recently been approved by the US Food and Drug Administration (FDA) for the treatment of rheumatoid arthritis and for sJIA, where it is remarkably effective for the treatment of both systemic and arthritic manifestations.

Aberrant IL-1 release has also been shown to have a role in many familial autoinflammatory disorders (see Case 33). Although levels of soluble IL-1 are difficult to measure in sJIA, this limitation may be technical. Serum from patients with sJIA has been shown to induce IL-1 secretion as well as the transcription of various innate immune genes from peripheral blood mononuclear cells from healthy individuals. IL-1 blockade can be highly effective in sJIA. In new-onset febrile disease, utilization of the recombinant IL-1 receptor antagonist anakinra results in steroid-free remission in more than half of the patients, and is therefore a common first-line therapeutic agent. The anti-IL-1β antibody canakinumab has also shown striking efficacy in randomized controlled trials, and is FDA approved for sJIA as well as for other conditions of IL-1 excess. The optimal therapeutic strategy for utilization of IL-1 versus IL-6 blockade in sJIA remains to be determined.

Questions.

1 *Why is sJIA considered to be primarily an autoinflammatory syndrome, rather than simply an arthritis?*

2 *What might be some possible side effects of IL-1 or IL-6 inhibition?*

3 *What other conditions involve IL-1 dysregulation?*

4 *Why is Catherine's rash only present in association with fever?*